Jishnu Prasad

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Education

National Institute of Technology Karnataka (NITK Surathkal) Surathkal, India
B.Tech in Electrical and Electronics Engineering *CGPA:* ***7.18 / 10*** *Sep. 2024 – Present*

Experience

SPIRE (Signal Processing, Interpretation and REpresentation) Lab **Indian Institute of Science (IISc)**
Research Intern | Multimodal Retrieval Systems & LLM Evaluation *May 2026 – July 2026*

- Built a **multilingual (English & Kannada) multimodal RAG system** for banking forms and documentation, integrating **OCR and web scraping** with a **multi-agent retrieval framework** that dynamically routed queries across PDF, Markdown, Excel, and HTML sources.
- Implemented a **Knowledge-Augmented Generation (KAG)** architecture to overcome the limitations of traditional RAG on complex queries, improving context recall and faithfulness.
- **Fine-tuned a Qwen3-27B-Instruct model** as an answer-quality critic scoring **Accuracy, Answer Relevancy, and Faithfulness** to drive a **DeepThink feedback loop**.

HALE (Healthcare Analytics and Language Engineering) Lab **IT Dept., NITK**
Research Intern | ROS2, Kubernetes, Redis, RAG, Ansible, Raspberry Pi *Jan. 2025 – Present*

- Designed a multimodal data pipeline fusing **satellite/drone imagery** with SLM-based **RAG** over textual disaster reports, and built a real-time natural-language query interface for disaster-intelligence decision support.
- Engineered and operated a multi-node **Raspberry Pi k3s cluster** with **MetalLB and NGINX Ingress** for load-balanced, TLS-secured edge-inference deployment.
- Built a full observability stack (**Prometheus, Grafana, Loki, Promtail**) and deployed **MinIO** object storage for distributed model/image serving across the cluster.
- Automated reproducible provisioning of edge-inference workloads on Raspberry Pi 5 nodes using **Ansible**.

Projects

Caterpillar Autonomy Challenge — Sand Berm Construction Rover | *ROS2, Python, LiDAR, YOLOv8, SLAM*

- Ranked **Top 10 of 1,922 teams** (IIT Madras); collaborated in a multidisciplinary team to design and deploy an autonomous rover capable of navigating unstructured outdoor terrain under competition constraints.
- Developed a LiDAR-based obstacle detection pipeline using `/scan` data to identify boulders and generate real-time avoidance footprints, integrated with SLAM-based occupancy grid mapping.
- Fine-tuned a YOLOv8n model for crater detection (~92% mAP), enabling robust visual perception and dynamic avoidance in unstructured environments.
- Engineered a sensor-fusion and path-planning stack combining LiDAR, odometry, and occupancy grids for autonomous navigation in non-line-of-sight (NLOS) conditions on resource-constrained hardware (Raspberry Pi 5 + Hailo AI HAT).

LLM-Guided Autonomous Rover | *ROS2, Raspberry Pi 5, ESP32, React Native, Gemini, YOLO*

- Developed an autonomous rover using ROS2 and a Raspberry Pi 5, leveraging a webcam-based perception pipeline to capture images and offload decision-making to Gemini for navigation commands.
- Implemented a cloud-inference loop where captured frames are processed by Gemini to generate real-time movement instructions, enabling LLM-driven control without onboard heavy vision models.
- Re-architected motor control by offloading actuation to an ESP32, establishing WiFi-based communication between a mobile interface and rover for low-latency command execution.
- Built a React Native app to stream camera input, interface with Gemini APIs, and transmit navigation commands, with an optional local YOLO-based fallback for on-device obstacle avoidance.

Multi-Camera Live Sky Stitching System | *OpenCV, PyTorch, Homography, Raspberry Pi, ESP32*

- Designed a low-cost multi-camera system to capture and stitch real-time night sky feeds into a wide-field panoramic view using four directional cameras.
- Implemented real-time image stitching using OpenCV with homography-based alignment, preprocessing, and seamless panorama generation.
- Developed a synchronized multi-sensor pipeline integrating webcams, MPU6050 IMU, ESP32, and Raspberry Pi for camera calibration and orientation-aware stitching.

Other Systems Projects | *Go, ReactJS, Django, NATS JetStream, Redis*

- **Memora** (GitHub): Redis-compatible in-memory database in Go with RESP protocol support, TTL expiration, and snapshot persistence, sustaining **226K+ ops/sec** at **39 μs P50** latency.
- **ATLAS** (GitHub): Fault-tolerant telemetry and log-aggregation platform with AI-assisted log analysis, real-time dashboards, and a NATS JetStream messaging layer supporting mTLS-secured fleet orchestration.

Technical Skills

Languages: Python, C++, Go, JavaScript/TypeScript, MATLAB
Robotics & CV: ROS2, SLAM, LiDAR, YOLOv8, OpenCV, Simulink, Vision Transformers
ML / AI: PyTorch, TensorFlow, LangChain, RAG, Multimodal AI, LLM Fine-Tuning
Systems & Tools: Docker, Kubernetes, Redis, PostgreSQL, Ansible, Git, AWS, MCP, Kafka

Leadership & Activities

Amateur Astronomy Club (AAC) & SMILE Club, NITK *Chief Technical Coordinator*
Co-led technical Arduino-based astronomy projects, led events, publicity, and sponsorships, and founded several sub-teams to improve team efficiency.

Schneider Electric Go Green 2025 (Team EcoLogic) *Team Lead*
Led a team designing a decentralized solar-powered hydrogen and biogas system for sustainable rural energy and waste management.